CELL SURFACE TARGETING BY GAG-BINDING PEPTIDES

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ABSTRACT

Cell-Penetrating Peptides (CPPs) are short peptides that can enter cells by spontaneously crossing the plasma membrane, either by endocytosis or direct translocation through the lipid bilayer. Several natural CPPs correspond to the third helix of a highly conserved domain of homeoproteins, the so-called homeodomain. A GAG-binding sequence is found upstream of the CPP sequence in homeodomains, with different GAG specificity depending on the homeoprotein. For example, the homeodomain of the protein En2 encompasses a sequence interacting preferentially with HS while the homeodomain of Otx2 carries a disulfated CS-binding sequence[1].

GAGs are known to participate in the internalization of homeoprotein derived CPPs and we recently showed that chimeric peptides composed of a GAG-binding sequence and a CPP sequence have enhanced internalization in certain cell lines depending on their GAG expression[2]. In order to quantitatively assess cell surface accumulation mediated by GAGs we are currently developing a fluorescence-based assay with chimeric peptides composed of a GAG-binding sequence linked to a fluorogenic cell-surface protease peptide substrate. We are using the rate of substrate cleavage as a reporter of cell surface accumulation through GAGs.

Ultimately, this will allow us to identify efficient GAG-binding sequences in order to design new chimeric CPPs with enhanced cell selectivity an internalization properties.

REFERENCES

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